

| Question | Marking Guidance                                                                                                                                                                                                                                         | Mark  | Comments                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01.1     | 1. High blood/hydrostatic pressure;<br>2. <b>Two named</b> small substances pass out eg water, glucose, ions, urea;<br>3. (Through small) gaps/pores/fenestrations <b>in</b> (capillary) endothelium;<br>4. (And) through (capillary) basement membrane; | 3 max | Ignore references to podocytes<br>1. Ignore references to afferent and efferent arterioles<br>1. Ignore 'increasing/higher blood pressure' as does not necessarily mean high<br>2. Accept correct named ions<br>2. Accept mineral ions/minerals<br>2. Accept amino acids/ <u>small</u> proteins<br>2. Ignore references to molecules <b>not</b> filtered<br>3. Accept epithelium for endothelium |
| 01.2     | <input checked="" type="checkbox"/> Glucose by facilitated diffusion and active transport and water down a water potential gradient                                                                                                                      | 1     |                                                                                                                                                                                                                                                                                                                                                                                                  |
| 01.3     | 17.4;                                                                                                                                                                                                                                                    | 1     | Accept any number of fours after the decimal point.                                                                                                                                                                                                                                                                                                                                              |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |                                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01.4 | <p>1. Thicker medulla means a longer loop (of Henle);</p> <p>2. (The longer the loop of Henle means) <b>increase</b> in sodium ion concentration (in medulla)</p> <p><b>OR</b></p> <p>(The longer the loop of Henle means) sodium ion gradient maintained for <b>longer</b> (in medulla)</p> <p><b>OR</b></p> <p>(The longer the loop of Henle means) <b>more</b> sodium ions are moved out (into medulla);</p> <p>3. (Therefore) water potential gradient maintained (for longer), <b>so</b> more water (re)absorbed (from loop and collecting duct);</p> <p><b>OR</b></p> <p>More water is (re)absorbed <b>from</b> the loop (of Henle) / collecting duct <b>by</b> osmosis;</p> | 3 | <p>2. Must have idea of increase/longer/more</p> <p>3. Reject water being reabsorbed <b>into</b> the loop of Henle</p> <p>3. Direction is important</p> <p>3. Accept <math>\Psi</math> for water potential</p> |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Question | Marking Guidance | Mark | Comments |
|----------|------------------|------|----------|
|----------|------------------|------|----------|